

Basics of Multivariate Statistics

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Statistics: everybody uses it, nobody understands it

Why Should you care?

We do Machine Learning not Statistic after all

Definition of Statistics

Statistics (from German: Statistik, orig. "description of a state, a country") is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data

Why is it still important?

- to organize, interpret and understanding data
- to build, select and evaluate models
- contains the building blocks of data science and data analysis
- math at the base of linear regression (neuron)
- powerful (and dangerous) tool to support arguments

- Basics
- Dataset, Mean and Variance
- Least Squared
- Linear Regression
- Logistic Regression
- Bias vs Variance

Basic tools for today

Probability Space

$(\Omega, \sigma(\Omega), \mathbb{P})$

- Ω set of all possible single outcomes
- $\sigma(\Omega)$ set of events (all possible combinations of single outcomes)
- $\mathbb{P}$ probability function

Random Variable

A Random Variable $X : \Omega \rightarrow E$ takes as input an outcome and returns a realization

- $(\Omega, \sigma(\Omega), \mathbb{P})$ probability space
- $(E, \mathcal{E})$ measurable space
- $\mathbb{P}(X(\omega) = x)$ is the probability that the realization x of the RV X occurs

Example: Rolling two dice

- $\Omega = \{(1, 1), (1, 2), \dots, (3, 5), \dots, (6, 6)\}$ all possible single events
- $E = \{2, 3, \dots, 12\}$ all possible single realizations
- $X(\omega) = \omega_1 + \omega_2$ is the sum of the two dice
- $\mathbb{P}(X = x) = \begin{cases} 1/36 & \text{for } x = 2, 12 \\ 2/36 & \text{for } x = 3, 11 \\ \vdots & \\ 1/6 & \text{for } x = 7 \end{cases}$

Basic Tools for today

Assume from here both Ω and E are of the form $\mathbb{R}^n$

Probability Density Function

a p.d.f. is a non negative Lebesgue-integrable function

$\mathbb{P}\{a \leq X \leq b\} = \int_a^b f_X(x) dx$ (think of it as the probability in the continuous case)

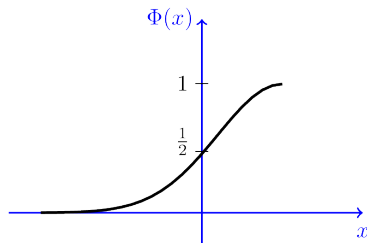
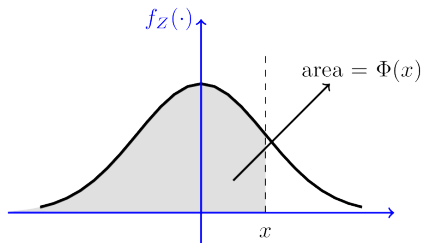
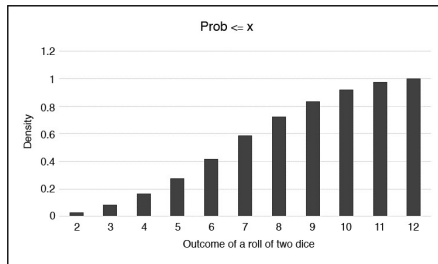
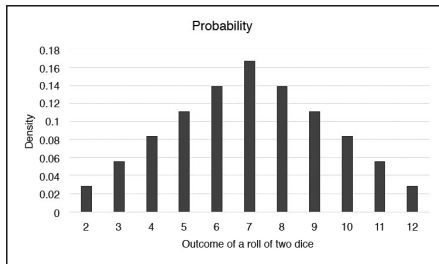
- $\int_{\mathbb{R}^n} f_X(x) dx = 1$
- $f_X(x) \geq 0 \quad \forall x$

Cumulative Density Function

a c.d.f is $F_X(x) = \int_{-\infty}^x f_X(t) dt = \mathbb{P}(X \leq x)$ (think of it as the sum of the probability of all events up to x)

- $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $\lim_{x \rightarrow +\infty} F_X(x) = 1$
- $f_X(x)$ is right continuous (only look at the limit definition from the right)

Basic Tools for today



Expected Value and Sample Mean

$$\mathbb{E}[X] = \int_{\mathbb{R}^n} xf(x)dx$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Variance and Sample Variance

$$\text{VAR}[(X - \mathbb{E}[X])^2] = \int_{\mathbb{R}^n} (x - \mathbb{E}[X])^2 f(x)dx$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Basic Tools for today

Covariance and Sample Covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \int_{\mathbb{R}^n} (x - \mathbb{E}[X])(y - \mathbb{E}[Y]) f(x, y) dx$$

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Correlation Coefficient and Sample Correlation

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

$$r_{XY} = \frac{S_{XY}}{S_X S_Y}$$

Theorem (Central Limit Theorem)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with mean μ and variance $\sigma^2 < \infty$. Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, as $n \rightarrow \infty$,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

Let's focus on data, how do we represent it?

Matrix representation

$$D = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} \quad x_{ij} : \text{value of variable } j \text{ in observation } i$$

Row perspective (observations):

$$\underline{x}_i^T = (x_{i1}, x_{i2}, \dots, x_{ip}) \quad \forall i \in \{1, \dots, n\}$$

Column perspective (all values of a covariate):

$$\underline{c}_j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix} \quad \forall j \in \{1, \dots, p\}$$

Consider the **Mean vector**

$$\underline{\bar{X}} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{pmatrix}$$

where the single elements are the mean of the columns, the mean of the covariates

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij} = \frac{\underline{1}^\top \underline{c}_j}{\underline{1}^\top \underline{1}} \underline{1}$$

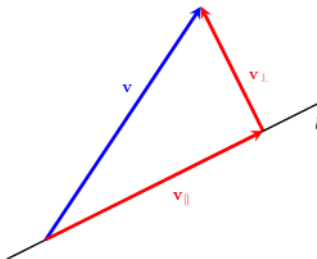
Think of this: any realization of a covariate (value of a variable) can be written as the sum of the mean plus whatever is left

$$\underline{c}_1 = \bar{x}_1 \underline{1} + \underline{d}_1$$

Something neat about $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$ is that it can be written as

$$\Pi_{c_1 | \mathcal{L}(\underline{1})} = \frac{\underline{1}^\top \underline{c}_1}{\underline{1}^\top \underline{1}} \underline{1}$$

which is the projection of the vector on the linear space where all elements are the same



Now we can continue this by writing out $\underline{d}_j$ as

$$\underline{d}_1 = \underline{c}_1 - \bar{x}_1 \underline{1} = \begin{pmatrix} x_{11} - \bar{x}_1 \\ x_{21} - \bar{x}_1 \\ \vdots \\ x_{n1} - \bar{x}_1 \end{pmatrix}$$

we notice that the 2-norm of $\underline{d}$ closely resembles the standard error (square root of variance)

$$\|\underline{d}_1\|_2 = \sqrt{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \propto \sqrt{s_{11}}$$

Main idea

any dataset can be thought of as the merge of two Linear Subspaces:

- the space of the Mean, of dimension 1, where everything is equal
- the space of Variance, of dimension $(n-1)$, orthogonal to the space of the Mean

Ever wondered why the formula of sample variance is divided by $(n-1)$ and not n ? This is why

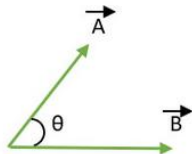
Datasets

Let's consider the interaction between two covariate vector $\underline{c}_1$ and $\underline{c}_2$

$$\underline{c}_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}, \quad \underline{c}_2 = \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix}$$

we can write $\underline{d}_1$ and $\underline{d}_2$ explicitly and represent them geometrically (like you can with any vector):

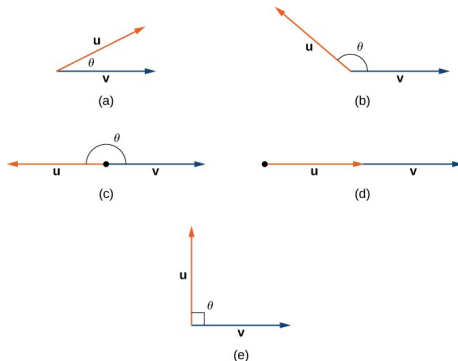
$$\underline{d}_1 = \underline{c}_1 - \bar{x}_1 \underline{1}, \quad \underline{d}_2 = \underline{c}_2 - \bar{x}_2 \underline{1}$$



If we remember how to calculate the cosine of an angle with vectors, we find something interesting

$$\begin{aligned}\cos \theta &= \frac{\langle d_1, d_2 \rangle}{\|d_1\| \|d_2\|} = \frac{d_1^\top d_2}{\sqrt{d_1^\top d_1} \sqrt{d_2^\top d_2}} \\&= \frac{\frac{1}{n} \sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)}{\sqrt{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \sqrt{\sum_{i=1}^n (x_{i2} - \bar{x}_2)^2}} \\&= \frac{s_{12}}{s_1 s_2} = \rho_{12}\end{aligned}$$

So now we see that the correlation coefficient is nothing more than the cosine of the angle between two covariates!



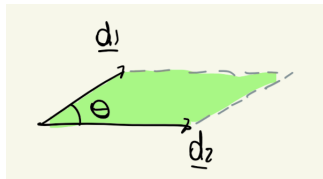
- c) covariates are the same but rescaled and opposite direction ($\rho = -1$)
- d) covariates are the same rescaled ($\rho = 1$)
- e) covariate are uncorrelated ($\rho = 0$)

Consider the **sample covariance matrix** for a simple case with two covariates

$$\begin{aligned} S &= \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix} = \frac{1}{n-1} \begin{bmatrix} d_1' d_1 & d_1' d_2 \\ d_2' d_1 & d_2' d_2 \end{bmatrix} \\ &= \frac{1}{n-1} \begin{bmatrix} \|d_1\|^2 & \|d_1\| \|d_2\| \cos \theta_{12} \\ \|d_1\| \|d_2\| \cos \theta_{12} & \|d_2\|^2 \end{bmatrix} \end{aligned}$$

- the **generalized variance** can be represented by the determinant $\det(S)$
- the **total variance** can be represented by the trace $tr(S) = s_{11} + \dots + s_{pp}$

generalized variance



$$\begin{aligned}\det S &= \frac{1}{(n-1)^2} \left(\|\underline{d}_1\|^2 \|\underline{d}_2\|^2 - \|\underline{d}_1\|^2 \|\underline{d}_2\|^2 \cos^2 \theta \right) \\ &= \frac{1}{(n-1)^2} \|\underline{d}_1\|^2 \|\underline{d}_2\|^2 \sin^2 \theta \propto \text{Area}(\text{parallelogram}(\underline{d}_1, \underline{d}_2))\end{aligned}$$

generalized variance

- If S_{11} increases $\Rightarrow \|d_1\|^2$ increases $\Rightarrow \det S$ increases
- If $\theta_{12} = 0 \Rightarrow \det S = 0 \Rightarrow$ there is no variability between the covariates
- If $\theta_{12} = \frac{\pi}{2} \Rightarrow \det S = \max$ if I don't change S_{11} and S_{22}

total variance Imagine adding a covariate to the mix

- the total variance will increase because the trace will get a new element $tr(S) = s_{11} + \dots + s_{pp} + s_{p+1,p+1}$

You might be thinking: "Mh, we haven't talked about probability at all in this"

And you'd be correct

That's because I lied to you



90% of Statistics is Linear Algebra, 10% is Probability



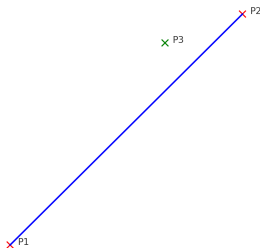


And that's why we need to talk about Linear Algebra just a little more

Break time! (5 minutes)

Least Squares

Least Squares Problem: What do we do when we try to have too many points to fit a line?



There's no line that goes through all points, so we need to come up with something else

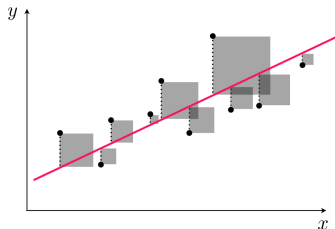
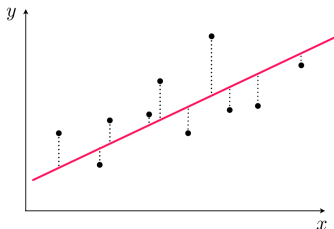
Least Squares

Let's work on a 2D plane first

Think of a point having coordinates $(x,y)=(a, b)$.

A line having the expression $b = \beta_0 + \beta_1 a$.

Our objective is going to be to find the line that minimizes the sum of the distances of the points from the line



Least Squares

Let's rewrite all our points in matrix form

$$A = \begin{bmatrix} 1 & a_{11} \\ 1 & a_{21} \\ \vdots & \vdots \\ 1 & a_{m1} \end{bmatrix}, \quad \underline{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \underline{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Our minimization of the squared distances can now be expressed like this!

$$\min_{(\beta_0, \beta_1)} \|Ax - b\|_2^2 = \min_{(\beta_0, \beta_1)} \sum_{i=1}^m (a_{i1}\beta_0 + a_{i2}\beta_1 - b_i)^2$$

How do we solve this?

Least Squares

Like we would any minimization problem, we take the derivative and set it to zero

$$\nabla_{\underline{\beta}} f(\underline{\beta}) = 0$$

Compute the gradient:

$$f(\underline{\beta}) = (\underline{A}\underline{\beta} - \underline{b})^\top (\underline{A}\underline{\beta} - \underline{b})$$

$$\nabla_{\underline{\beta}} f(\underline{\beta}) = 2\underline{A}^\top (\underline{A}\underline{\beta} - \underline{b})$$

Set it to zero:

$$\underline{A}^\top (\underline{A}\underline{\beta} - \underline{b}) = 0$$

Let's now rewrite to get the full solution for the coefficient of our Least Squares Line

$$A^{\top}(A\underline{\beta} - \underline{b}) = 0 \quad \rightarrow \quad A^{\top}A\underline{\beta} = A^{\top}\underline{b}$$

$$\underline{\beta} = (A^{\top}A)^{-1}A^{\top}\underline{b}, \quad \text{if } A^{\top}A \text{ is invertible}$$

Why do we need this?

Linear Regression

Linear Regression

Setup:

We assume that our variable of interest can be expressed as a linear combination of the other covariates, plus an error.

Observation-wise form:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n$$

Matrix form:

$$\underline{y} = \mathbb{X} \underline{\beta} + \underline{\varepsilon}$$

where:

- $\underline{y}$: $n \times 1$ vector of **responses**
- $\mathbb{X}$: $n \times (p + 1)$ **design matrix** with first column of ones
- $\underline{\beta}$: $(p + 1) \times 1$ vector of **coefficients**
- $\underline{\varepsilon}$: $n \times 1$ **error** vector with variance $\sigma^2 I$

$$\underline{y} = \mathbb{X} \underline{\beta} + \underline{\varepsilon}$$

Remarks:

- 1 $E[\underline{\varepsilon} \mid \mathbb{X}] = \underline{0}$ (zero mean, exogeneity)
- 2 $\text{Var}(\underline{\varepsilon} \mid \mathbb{X}) = \sigma^2 I$ (homoscedasticity, no autocorrelation)
- 3 $\text{rank}(\mathbb{X}) = p + 1$ (no perfect collinearity)
- 4 usually for simplicity it's assumed that $\underline{\varepsilon} \sim \mathcal{N}(\underline{0}, \sigma^2 I)$

Linear Regression

$$\underline{\hat{y}} = \mathbb{X}\underline{\hat{\beta}}, \quad \underline{\hat{\varepsilon}} = \underline{y} - \underline{\hat{y}}$$

Estimate of σ^2 :

$$\hat{\sigma}^2 = \frac{\underline{\hat{\varepsilon}}^T \underline{\hat{\varepsilon}}}{n - (p + 1)}$$

Standard error of $\hat{\beta}_j$:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbb{X}^T \mathbb{X})^{-1}]_{jj}}$$

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{RSS} = \sum_{i=1}^n \hat{\varepsilon}_i^2, \quad \text{ESS} = \text{TSS} - \text{RSS}$$

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

How do we solve this problem to find the coefficients? With Least Squares, we are still fitting a line

Solution of OLS (Ordinary Least Squares)

$$\hat{\underline{\beta}} = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \underline{y}$$

Key Idea: the reason why Statistics exists is because of $\underline{\varepsilon}$. It is a Random Variable, and it is responsible for introducing stochasticity (randomness)

Because $\underline{\varepsilon}$ is a RV, so is $\underline{y}$, and by consequence so is $\underline{\hat{\beta}}$

$$\mathbb{E}[\underline{\hat{\beta}} \mid \mathbb{X}] = \underline{\beta} \quad (\text{unbiased})$$

$$\text{Var}(\underline{\hat{\beta}} \mid \mathbb{X}) = \sigma^2 (\mathbb{X}^T \mathbb{X})^{-1}$$

It is also important to take a look at the **fitted values** (or predicted values)

$$\hat{\underline{y}} = \mathbb{X}\hat{\underline{\beta}}$$

And the difference between the real values and our predicted ones, called **residuals**

$$\hat{\underline{\varepsilon}} = \underline{y} - \hat{\underline{y}}$$

$$\underline{\hat{\varepsilon}} = \underline{y} - \underline{\hat{y}}$$

From these we can deduce:

Estimate of σ^2 :

$$\hat{\sigma}^2 = \frac{\underline{\hat{\varepsilon}}^T \underline{\hat{\varepsilon}}}{n - (p + 1)}$$

Standard error of $\hat{\beta}_j$:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbb{X}^T \mathbb{X})^{-1}]_{jj}}$$

let's look at an example together

Linear Regression

Dataset "mpg" on data of different models of cars

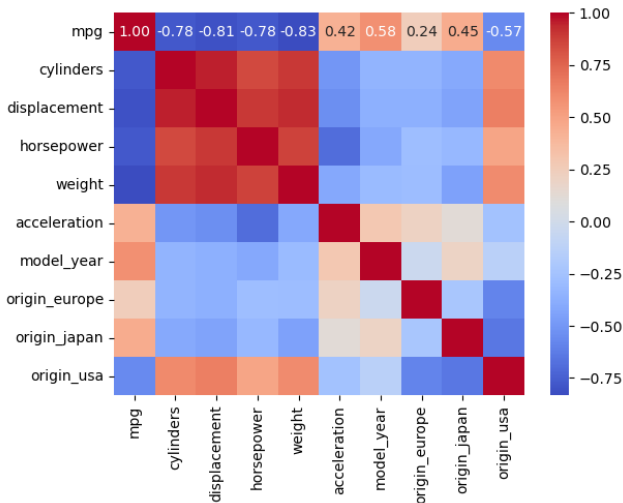
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as smf
import scipy.stats as stats

mtcars = sns.load_dataset("mpg").dropna()
mtcars.head()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino

Linear Regression

first, let's look at the correlation matrix



Linear Regression

This is what the linear regression function prints

```
model = smf.ols("mpg ~ weight", data=mtcars).fit()
print(model.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          mpg      R-squared:          0.693
Model:                  OLS      Adj. R-squared:      0.692
Method:                 Least Squares      F-statistic:    878.8
Date:                   Fri, 19 Sep 2025    Prob (F-statistic): 6.02e-102
Time:                   14:43:52      Log-Likelihood:   -1130.0
No. Observations:      392      AIC:                2264.
Df Residuals:          390      BIC:                2272.
Df Model:               1
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	46.2165	0.799	57.867	0.000	44.646	47.787
weight	-0.0076	0.000	-29.645	0.000	-0.008	-0.007

```

=====
Omnibus:                 41.682      Durbin-Watson:      0.808
Prob(Omnibus):            0.000      Jarque-Bera (JB):    60.039
Skew:                     0.727      Prob(JB):            9.18e-14
Kurtosis:                 4.251      Cond. No.            1.13e+04
=====
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.13e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Linear Regression

It's a lot, but we only care about a few things

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1. Significance of the coefficients

A really handy relationship that evolves from this is that you can create a test statistic that has a known distribution

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}$$

If we assume that one specific coefficient is 0, then

$$H_0 : \beta_j = 0 \rightarrow t_j \sim t_{n-(p+1)} \quad (\text{t-student distribution})$$

Confidence interval: if 0 is contained in this interval, it means that there is not enough statistical evidence to be able to confirm that β_j couldn't actually be 0

$$CI_{\alpha}(\beta_j) = \left[\hat{\beta}_j \pm t_{\alpha/2, n-(p+1)} \cdot SE(\hat{\beta}_j) \right]$$

If 0 is outside of the interval, then we can reject the hypothesis that β_j could have been 0 to begin with. In other words we are saying that we are fairly certain that β_j isn't 0, therefore significant to our model

They have an associated **p-value**, which we can think of as the probability of being wrong if we keep the coefficient in the model

$$p\text{-value}_j = 2 \mathbb{P}\left(T_{n-(p+1)} > |t_j|\right)$$

We generally look for p-values smaller than 5%

2. Normality of the model and qqplot

We take the residuals, order them, and standardize them into a distribution $N(0, 1)$. To do that, we calculate their z counterpart that we know is distributed like a Standard Normal because of properties of the Normal distribution.

$$e_i = y_i - \hat{y}_i$$

$$r_i = \frac{e_i}{\hat{\sigma} \sqrt{1 - h_{ii}}}$$

$$\hat{\sigma}^2 = \frac{1}{n - p - 1} \sum_{j=1}^n e_j^2$$

$$H = X(X^\top X)^{-1}X^\top, \quad h_{ii} = \text{the } i\text{-th diagonal element of } H$$

Linear Regression

This is what the linear regression function print

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OLS Regression Results

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Linear Regression

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Bibliography https://en.wikipedia.org/wiki/Probability_space[https :
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//en.wikipedia.org/wiki/Random_variable](https://en.wikipedia.org/wiki/Random_variable)[https :
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